

Vtemp Extraction & Comparison Flow ($f_{\min}$)

1. Testbench Configuration

Testbench: PrimeSim.default.VtempReal

Design: Tese/Var_Comp/schematic.VtempReal

Analyses:

- **AC Analysis (ac):** Point of Interest at 6 GHz ($f_{\min}$)
- **DC Analysis (dc_op / dc):** Operating point extraction

Sweep: Temperature sweep (temp) from -40°C to 125°C (step 16.5°C) integrated into the corner setup.

Corners: TT (Typical), FF (Best-Case Q / Bcq), SS (Worst-Case Q / Wcq) with tandem temperature sweeps.

2. Output Expressions

OUTPUT	EXPRESSION
I_{dev} (Device_current)	i(/V8/PLUS)
$C_{\text{max,meas}}$	abs(imag(Device_current) / (2 * 3.14159 * 6e9))
$C_{\text{max,cont}}$	flatten(Cap_max_measured, temp)
Q-factor	abs(imag(Device_current) / real(Device_current))
ΔT (C_{max})	xmax(Cap_max_measured_cont) - xmin(Cap_max_measured_cont)
ΔC (C_{max})	ymax(Cap_max_measured_cont, -40, 125) - ymin(Cap_max_measured_cont, -40, 125)
d_{CT} (C_{max})	Delta_C_cap_max / Delta_Temp_cap_max
$C_{\text{std,max}}$	ymax(Cap_max_measured_cont, -40, 125) / Cap_max_measured_cont
% Error $C_{\text{std,max}}$	(ymax(C_standard_cap_max, -40, 125) - ymin(C_standard_cap_max, -40, 125)) * 100
$V_{\text{CO,DC}}$	v(/net65)
V_{temp}	v(/Vtemp)